MAR ATHANASIUS COLLEGE OF ENGINEERING
KOTHAMANGALAM

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

ANALOG CIRCUITS

MICRO PROJECT REPORT

ON

WATER LEVEL INDICATOR USING BC547 TRANSISTOR

Submitted by:

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ACKNOWLEDGEMENT

We express our heartfelt gratitude to everyone who has played a vital role in supporting and contributing to the successful completion of our project. Firstly, we are thankful to God Almighty for His blessings, as it is through His grace that we accomplished our project.

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TITLE: WATER LEVEL INDICATOR

AIM

The aim of this project is to design and implement a water level indicator using transistors to monitor and display four different water levels using LEDs. A buzzer is incorporated to alert when the water reaches the maximum level, ensuring efficient water management and preventing overflow.

CIRCUIT DESIGN

Components:

1. Breadboard
2. NPN Transistors (BC547)
3. LED indicators
4. Buzzer
5. Resistors
6. Connecting wires
7. Power supply (9V Battery)

Circuit Description:

The Water Level Indicator circuit uses BC547 NPN transistors to detect and display water levels with LEDs and a buzzer. A 9V battery powers the circuit. Sensing probes placed at four levels inside the water tank act as inputs. As water reaches each level, it completes the circuit, providing a small base current to the corresponding BC547 transistor, turning it ON. This allows current to flow from the collector to the emitter, lighting up the respective LED.

Each LED is connected in series with a 1kΩ resistor to limit current. The final stage includes a buzzer, which activates when the highest level is reached, providing an audible overflow alert. The circuit operates in a simple switching mode, making it an efficient and cost-effective water level monitoring system.

WORKING PRINCIPLE

1. No Water Condition (All LEDs OFF)

When there is no water in the tank, all the transistors are in cutoff mode, meaning they are OFF. No base current flows, so the collector-emitter path remains open, and all LEDs and the buzzer remain OFF.

2. Water Reaches the First Level

When the water level rises and touches the first sensing probe, a small current flows through the 10kΩ resistor to the base of the first BC547 transistor. This causes the transistor to enter saturation mode (ON), allowing current to flow from the collector to the emitter. As a result, the first LED lights up, indicating the first water level.

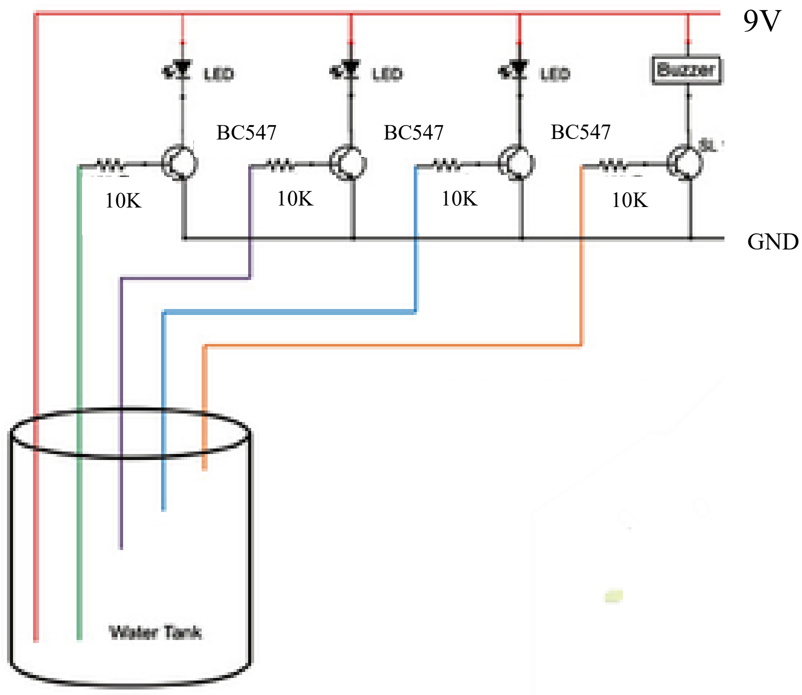
3. Water Reaches the Second, Third, and Fourth Levels

As water continues to rise, it successively touches the second, third, and fourth probes. Each corresponding transistor gets activated, turning ON its respective LED. This provides a visual representation of the water level in the tank.

4. Water Reaches the Maximum Level (Buzzer Activation)

When the water reaches the highest level, the last probe triggers the final transistor. This turns ON both the last LED and the buzzer. The buzzer serves as an audible warning to indicate that the tank is full, preventing overflow.

CIRCUIT DIAGRAM



CIRCUIT PARAMETERS

1. Power Supply (9V Battery):

The circuit is powered by a 9V battery, providing the required voltage for the transistors, LEDs, and buzzer.

2. Water Level Probes (Sensors):

- The sensing wires are placed at four different levels inside the water tank.
- Each probe corresponds to a specific water level, and the last probe (highest level) also activates a buzzer.

3. NPN Transistors (BC547) as Switches:

- The BC547 transistors are used in switching mode (ON/OFF) to control LEDs and the buzzer.
- The **base** of each transistor is connected to a sensing probe through a **10kΩ resistor**, limiting base current.
- The **collector** is connected to the LED (and buzzer for the last stage), while the **emitter** is connected to ground.

4. LEDs with Current-Limiting Resistors (1kΩ):

- Each LED is connected in series with a **1kΩ resistor** to protect it from excessive current.
- The LEDs light up sequentially as water reaches each level.

5. Buzzer for Overflow Alert:

- The highest water level probe is connected to the base of the last transistor.
- When water reaches this level, the transistor turns ON, activating both an LED and a buzzer. This provides an audible alert to signal a full tank.

SIMULATION AND TESTING

Simulate in Tinkercad

Build on Breadboard: Assemble the circuit on a breadboard, connecting each component as per the diagram

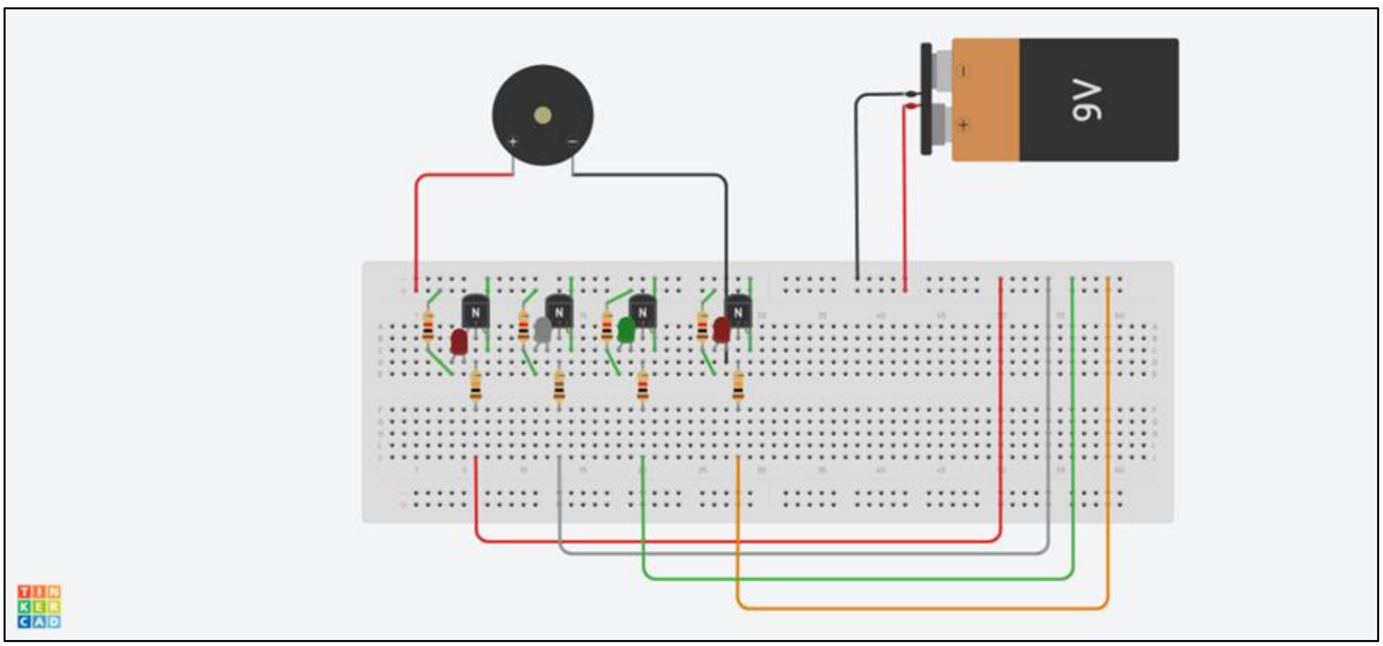
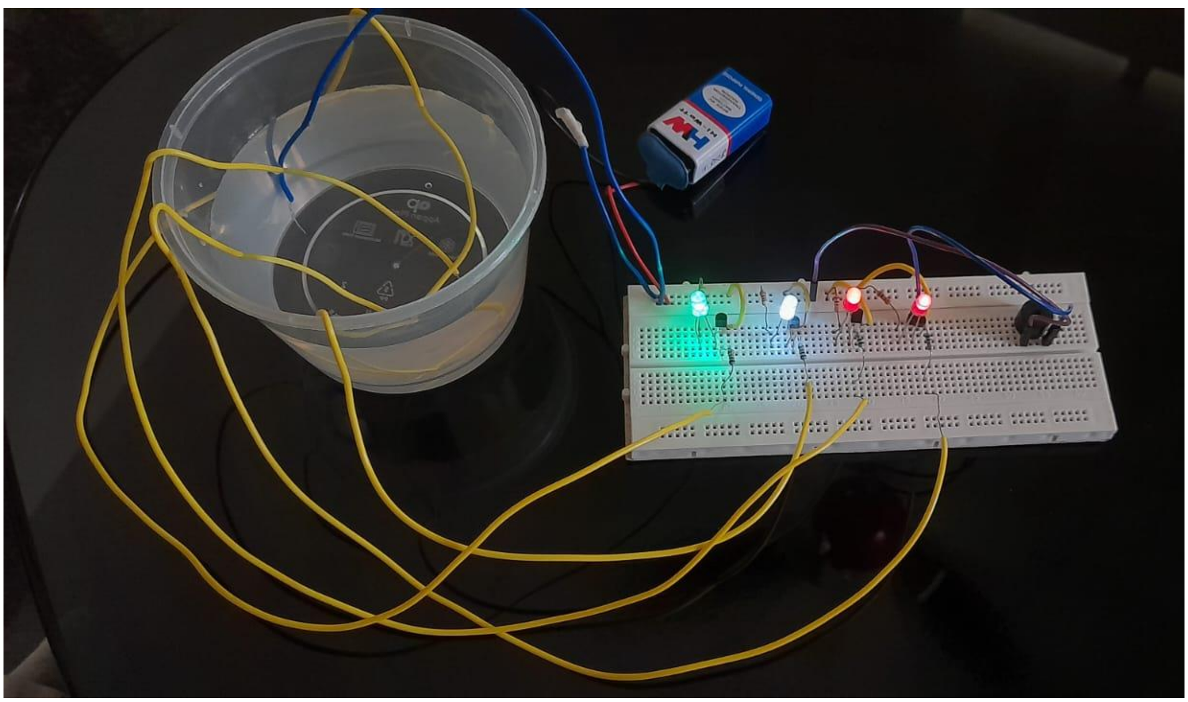


FIG 1.2: TINKERCAD SIMULATION

The circuit was simulated in Tinkercad to verify its functionality before hardware implementation. In the absence of water in simulation, the probes from base of transistors were manually connected to the positive rail (9V) to simulate water conductivity. When each probe was connected, the corresponding BC547 transistor switched ON, allowing current flow and lighting up the respective LED. The LEDs were tested to ensure they turned ON sequentially as water levels increased. Finally, the last probe activated both the LED and the buzzer, confirming the overflow alert. The simulation successfully validated the circuit's operation before breadboard assembly.

HARDWARE IMPLEMENTATION

The water level indicator circuit was assembled on a breadboard. The sensing probes were connected at different levels inside the tank. As water reached each level, the respective transistor switched ON, lighting up the corresponding LED. At the highest level, the buzzer activated, providing an overflow alert. The circuit was tested to ensure proper LED sequencing and buzzer operation, confirming successful hardware implementation. Below is the image of the assembled circuit on the breadboard.



RESULTS

The Water Level Indicator circuit was successfully simulated in TinkerCAD and implemented on a breadboard. During testing, the circuit functioned as expected:

- LEDs lit up sequentially as water levels increased, accurately indicating the water level.
- The buzzer activated only at the highest level, providing an effective overflow alert.
- The transistors operated correctly as switches, turning ON and OFF in response to simulated water contact.

The circuit proved to be reliable, responsive, and easy to implement, making it an effective solution for water level monitoring.

CONCLUSION

The Water Level Indicator successfully demonstrated the use of transistors as electronic switches for liquid level sensing, controlling LEDs for visual indication and a buzzer for an audible alert. This project highlights the practical applications of transistors in automation and sensing systems, commonly used in water level controllers, motor drivers, and industrial automation. This system provides a cost-effective and efficient solution for water management, helping to prevent overflow and water wastage. Further improvements, such as automatic motor control, could enhance its functionality for real-world applications.